

Lecture 15: The Six-Factor Formula and Prompt Kinetics

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Chapter 7:** Time-Dependent Reactor Physics.
 - Sections 7.1 – 7.2: Reactor Kinetics and Delayed Neutrons.
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1 The "Bridge": From Buckling to the Six-Factor Formula

Over the last week, we analyzed reactors in two separate parts:

1. **The Physics (k_∞):** The Four-Factor Formula ($\epsilon p f \eta$) describes an infinite medium.
2. **The Geometry (Buckling):** The balance between Material Buckling (B_m^2) and Geometric Buckling (B_g^2) describes leakage.

In the CP-1 case study, we saw that Fermi needed to minimize leakage to reach criticality. We can formalize this by defining the **Effective Multiplication Factor**, k_{eff} .

1.1 The Six-Factor Formula

For a finite reactor, we multiply k_∞ by the probability that neutrons do *not* leak out. Since neutrons exist in two distinct energy groups (Fast and Thermal), we add two factors:

$$k_{eff} = k_\infty \cdot P_{FNL} \cdot P_{TNL} \quad (1)$$

Expanding k_∞ :

$$k_{eff} = \epsilon p f \eta P_{FNL} P_{TNL} \quad (2)$$

1.2 Defining the Leakage Factors

Using the diffusion theory we derived:

1. Thermal Non-Leakage Probability (P_{TNL}): This is the probability a thermal neutron is absorbed in the core before diffusing out. From the one-group diffusion equation solution, this relates to the Geometric Buckling (B_g^2) and the Diffusion Length (L^2):

$$P_{TNL} = \frac{1}{1 + L^2 B_g^2} \quad (3)$$

Note: If the reactor is infinite, $B_g^2 \rightarrow 0$, so $P_{TNL} \rightarrow 1$.

2. Fast Non-Leakage Probability (P_{FNL}): Fast neutrons also migrate while slowing down. We define a new parameter, the ****Fermi Age (τ_T)****. Despite the name "Age," it has units of area (cm^2), just like L^2 . It represents $(1/6)$ of the mean square distance a neutron travels from birth (fission) to thermalization.

$$P_{FNL} = e^{-B_g^2 \tau_T} \quad (4)$$

Approximation: For large reactors (small B_g^2), $e^{-x} \approx 1 - x$ or $1/(1 + x)$.

2 Introduction to Reactor Kinetics

So far, we have assumed $k_{eff} = 1$ exactly (Criticality). What happens if $k_{eff} \neq 1$? The neutron population $n(t)$ changes with time.

2.1 The Reactivity (ρ)

We define ****Reactivity**** as the fractional departure from criticality:

$$\rho = \frac{k_{eff} - 1}{k_{eff}} = \frac{\Delta k}{k} \quad (5)$$

- $\rho = 0$: Critical ($k = 1$).
- $\rho > 0$: Super-critical.
- $\rho < 0$: Sub-critical.

3 Prompt Kinetics (The "Bomb" Approximation)

Let's assume for a moment that all neutrons are emitted instantly at the moment of fission. These are called ****Prompt Neutrons****.

3.1 The Kinetic Equation

The rate of change of the neutron population $n(t)$ is:

$$\frac{dn}{dt} = \text{Production} - \text{Loss}$$

$$\text{Production} = k_{eff} \frac{n}{l}, \quad \text{Loss} = \frac{n}{l}$$

Where l is the ****Mean Prompt Neutron Generation Time**** (lifetime of a prompt neutron from birth to death).

$$\frac{dn}{dt} = \frac{k_{eff}n}{l} - \frac{n}{l} = \frac{k_{eff} - 1}{l}n \quad (6)$$

3.2 The Exponential Excursion

Solving this differential equation ($n(t) = n_0 e^{t/T}$) with a period $T = \frac{l}{k-1}$: Let's plug in numbers for a thermal reactor.

- Mean Generation Time (l): $\approx 10^{-4}$ seconds.
- Reactivity: Suppose we are Supercritical by just 0.1% ($k = 1.001$).

$$T = \frac{10^{-4}}{0.001} = 0.1 \text{ seconds}$$

In just ****1 second**** ($t = 1$), the power rises by $e^{10} \approx 22,000$. **Conclusion:** If all neutrons were prompt, a nuclear reactor would be impossible to control. No human or machine could react fast enough.

4 Case Study: The Demon Core (1946)

The most famous example of Prompt Kinetics in action is the accident involving Louis Slotin at Los Alamos. This incident marks the transition from the "cowboy era" of nuclear physics to the modern era of remote safety.

4.1 The Context and Fermi's Warning

Slotin was the lead armorer at Los Alamos and was famous for his bravado in performing an experiment dubbed "Tickling the Dragon's Tail," where he would bring a bare sphere of Plutonium near criticality by hand to verify its readiness. Enrico Fermi famously warned Slotin and his team about this specific manual technique:

"If you keep doing that experiment, you will be dead in a year."

Fermi was tragically correct.

4.2 The Experiment (May 21, 1946)

- **The Core:** A 6.2 kg sphere of Plutonium-239 (The "Rufus" Core, later dubbed the Demon Core).
- **The Reflector:** Two half-spheres of Beryllium (a neutron reflector).
- **The Action:** Slotin held the top half of the reflector with his left hand and used a flathead screwdriver in his right hand to maintain a tiny gap, preventing the reflector from closing completely.

4.3 The Excursion

The screwdriver slipped. The upper Beryllium shell fell onto the core.

1. **Reflection:** The gap closed. Neutrons that were previously leaking ($P_{NL} < 1$) were suddenly reflected back.
2. **Prompt Criticality:** The system went ****Prompt Supercritical**** ($k_{eff} > 1 + \beta$).

3. **The Blue Flash:** Ionization of the air (and Cherenkov radiation in the eye fluid) caused a bright blue flash.
4. **The Reaction:** Slotin instinctively jerked his hand up, flipping the tamper off. This reaction took a fraction of a second.

4.4 The Result

- **Biological:** Even though the excursion was terminated by thermal expansion and Slotin's reaction ($\sim$ milliseconds), the prompt flux was so high that Slotin received ~ 21 Sv (2100 rem) of neutron/gamma radiation. He died 9 days later from acute radiation poisoning.
- **Material:** The burst produced $\sim 3 \times 10^{15}$ fissions. This created a massive inventory of fission products within the Plutonium sphere itself. The core became too radioactive ("hot") to be used as a weapon core for the upcoming Operation Crossroads tests. It had to be melted down and reprocessed to remove the contaminants.

4.5 The Protocol Shift

This accident ended the era of "hands-on" criticality experiments.

- **Immediate Ban:** Los Alamos immediately banned all manual manipulation of critical assemblies.
- **Remote Assembly:** All future criticality experiments were performed using remote-controlled machines (such as the "Lady Godiva" device) in shielded concrete bunkers (Kivas), with personnel located a quarter-mile away.

Lesson: In a prompt critical system, the "Period" T is shorter than human reaction time. Engineering controls (geometry, remote operation) must replace administrative controls (skill/procedure).

5 The Saving Grace: Delayed Neutrons

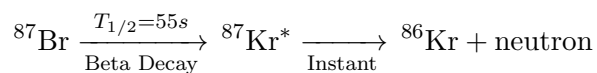
If prompt neutrons were the whole story, nuclear power would not exist. Fortunately, a small fraction of neutrons are **Delayed**.

5.1 The Mechanism: Precursor Decay

When U-235 fissions, it splits into two unstable, neutron-rich fragments. These fragments must undergo Beta Decay ($n \rightarrow p + e^- + \bar{\nu}$) to reach stability. Sometimes, a fragment is so unstable that after the beta decay, the daughter nucleus is highly excited and ejects a neutron directly to reach the ground state.

We call the parent fragment a **Precursor**. The neutron appears "delayed" not because the emission itself is slow, but because it had to wait for the Precursor to undergo beta decay first.

- **Example:** Bromine-87 is a fission product.



5.2 Quantifying the Delay

To understand the impact on the reactor, we look at two parameters: *How many* are delayed (β), and *how long* they wait (τ).

5.2.1 The Delayed Fraction (β)

We define β as the fraction of all fission neutrons that are delayed:

$$\beta = \frac{\text{Delayed Neutrons}}{\text{Total Neutrons}}$$

This value depends heavily on the fuel type, which dictates how "twitchy" (sensitive) a reactor is.

Isotope	Delayed Fraction (β)	Implication
Uranium-235	0.0065 (0.65%)	Standard Control Margin
Plutonium-239	0.0021 (0.21%)	Very Small Margin ("Twitchy")
Uranium-233	0.0026 (0.26%)	Small Margin

Table 1: Delayed Neutron Fractions for common fuels. Note that Pu-239 has $\approx 1/3$ the margin of U-235.

5.2.2 The Effective Time Constant

While ^{87}Br (55s) is the longest-lived precursor, there are dozens of others with half-lives ranging down to **0.2 seconds**.

- **Average Lifetime:** For U-235, the weighted average mean life of a delayed neutron is ≈ 12.7 seconds.
- *Note:* In the next lecture, we will break these down into 6 specific groups to derive the exact control equations.

5.3 The Consequence: Why Delay Creates Stability

The 12.7-second delay acts as a "Time Buffer" that allows both human operators and physical laws to control the reactor.

5.3.1 1. Mechanical Control (The Human Factor)

If the reactor period were determined by prompt neutrons (10^{-4} s), no machine could react fast enough to stop an excursion. The delayed neutrons slow the dominant time constant to seconds or minutes, allowing mechanical control rods to move and human operators to react.

5.3.2 2. Thermal Feedback (The Physics Factor)

Modern reactors rely on **Negative Temperature Coefficients** (e.g., Doppler Broadening) for inherent safety. However, this feedback is not instant—heat must transfer from the fuel to the coolant.

- **The Race Against Time:** Heat transfer takes seconds. Prompt excursions take milliseconds.
- **Without Delay:** The fuel would melt/vaporize before the heat could transfer to trigger the negative feedback.
- **With Delay:** The power rise is slow enough that the fuel heats up gradually, triggering the negative feedback and stabilizing the reactor automatically.

5.4 Defining Criticality Regimes

Because of this distinction, we define two very different states of "Criticality":

- **Delayed Critical ($k = 1$):** Steady state. The chain reaction relies on the delayed neutrons to keep going. Time scale = Seconds/Minutes. (Safe).
- **Prompt Critical ($k \geq 1 + \beta$):** The chain reaction sustains itself on prompt neutrons alone. The delayed neutrons are just "icing on the cake." Time scale = Milliseconds. (Explosive).

6 Summary

1. k_{eff} combines material properties (k_{∞}) and geometry (P_{NL}).
2. If $k_{eff} > 1$, power rises exponentially.
3. Based on prompt neutrons alone ($l \approx 10^{-4}$ s), the time scale is too fast to control.
4. **Next Lecture:** We will derive the "Inhour Equation" to see exactly how delayed neutrons (β) make nuclear power possible.